

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 3: Taylor's Expansion

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1 Keywords

- Taylor expansion, convergence, residue, gradient, tangent plane, Hessian, minima, maxima, saddle points

2 Main Points

2.1 Polynomial Approximation

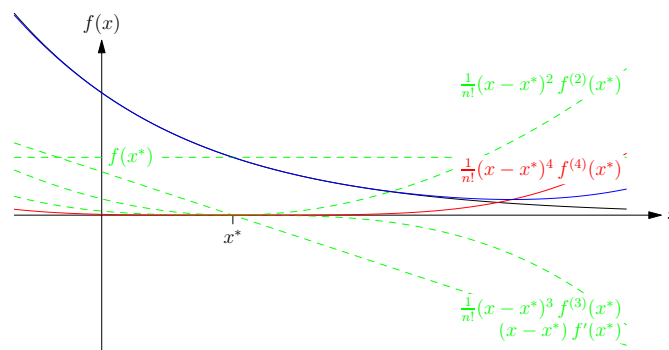
- Much of machine learning is about approximating functions, and every approximation scheme assumes something about how regular the function is
- A Taylor expansion approximates a smooth function near a point x^* by a polynomial

$$f(x) \approx a_0 + a_1(x - x^*) + a_2(x - x^*)^2 + \cdots + a_n(x - x^*)^n$$

- Differentiating both sides i times and setting $x = x^*$ picks out a single coefficient, so if we demand that the two sides agree in value and in their first n derivatives at x^* then $a_i = f^{(i)}(x^*)/i!$
- This gives the Taylor expansion

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 + \cdots + \frac{1}{n!}f^{(n)}(x^*)(x - x^*)^n$$

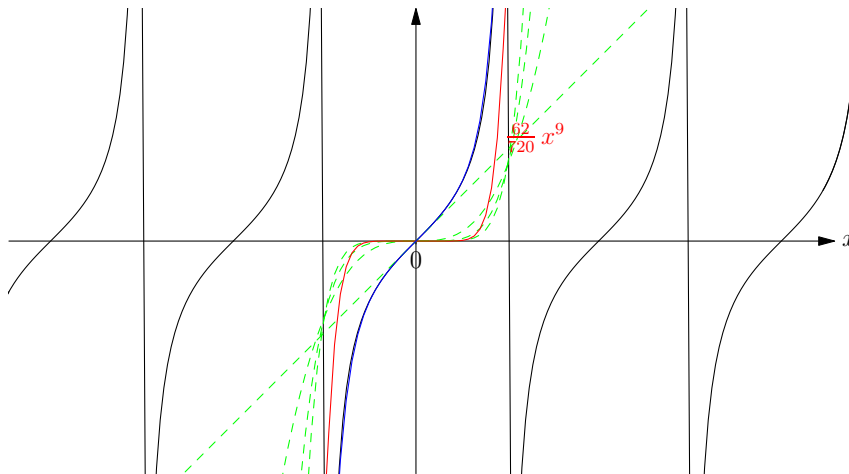
- The figure shows successive Taylor approximations to e^x around $x^* = 0$; each extra term extends the region where the approximation is good



2.2 Convergence

- Adding terms improves the approximation close to x^* , provided the higher derivatives exist
- Polynomials only diverge as $x \rightarrow \pm\infty$, so no polynomial can follow a function that blows up at a finite point, such as $1/x$, $1/(x - 5)^2$ or $\tan(x)$

- For such functions the Taylor series only converges in a limited region. The lecture animation builds up the expansion of $\tan(x)$ term by term; the final frame is below. However many terms we add, the series fails beyond the singularities at $x = \pm\pi/2$

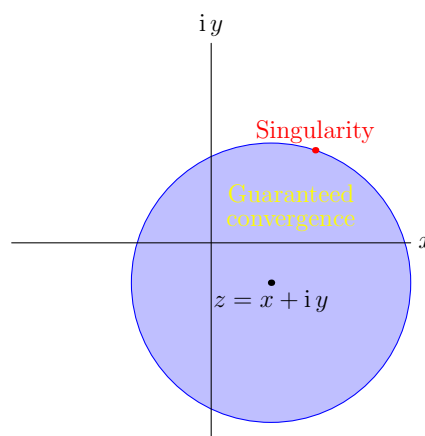


- To understand where a Taylor series converges we need to extend the function into the complex plane, $z = x + iy$
- A function is *analytic* where its derivative

$$\frac{df(z)}{dz} = \lim_{\epsilon \rightarrow 0} \frac{f(z + \epsilon) - f(z)}{\epsilon}$$

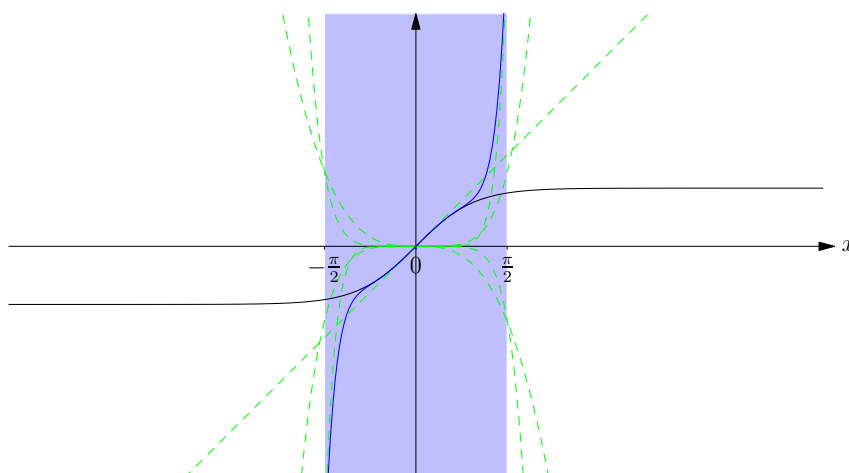
is the same whichever direction the complex number ϵ approaches zero from

- polynomials, exponentials, trigonometric functions, logarithms, ratios of polynomials and most "special functions" are analytic except at isolated singular points
- **Key result:** the Taylor series of an analytic function about z^* converges inside the largest circle centred on z^* that contains no singularity. The radius of that circle is the *radius of convergence*



- Singularities off the real axis matter even when the function looks perfectly well behaved for real x
 - $f(z) = 1/(z + i)^2$ is smooth for every real x , but it is singular at $z = -i$, so its Taylor series about 0 only converges for $|x| < 1$
- **The hyperbolic tangent.** $\tanh(x)$ has no singularities on the real line. But $\tan(ix) = i \tanh(x)$, and $\tan$ is singular at $\pi/2 + k\pi$, so $\tanh$ is singular at $z = \pm i\pi/2, \pm 3i\pi/2, \dots$ on the imaginary axis

- The expansion of $\tanh(x)$ about 0 therefore converges only for $|x| < \pi/2$, even though $\tanh$ itself is bounded everywhere



2.3 The Residue

- A Taylor expansion comes with an exact error term. Stopping after n terms,

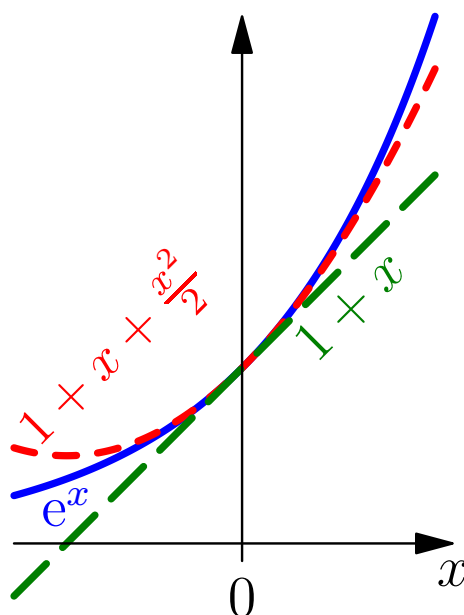
$$f(x) = \sum_{i=0}^{n-1} \frac{(x - x^*)^i}{i!} f^{(i)}(x^*) + \frac{(x - x^*)^n}{n!} f^{(n)}(\theta_x)$$

where θ_x is some number between x^* and x

- This last term, the *residue* (or remainder), is very useful for proving inequalities, because we often know the sign of $f^{(n)}$ even when we don't know θ_x
- Expanding e^x about 0,

$$e^x = 1 + x + \frac{x^2}{2} e^{\theta_x}$$

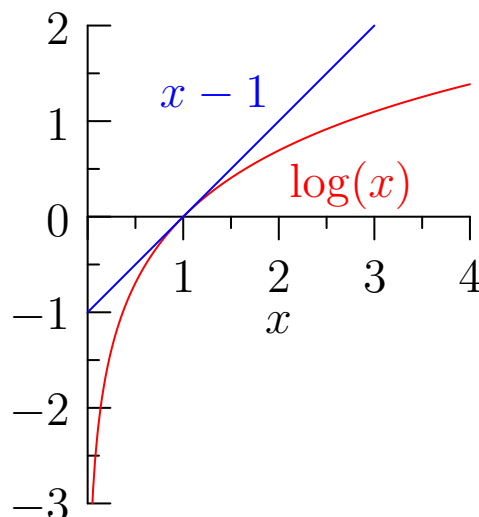
and since $e^{\theta_x} > 0$ we get the much-used bound $e^x \geq 1 + x$ for all x



- Expanding $\log(x)$ about $x = 1$,

$$\log(x) = (x - 1) - \frac{(x - 1)^2}{2\theta^2}$$

so $\log(x) \leq x - 1$, or equivalently $\log(1 + x) \leq x$



- The proof of the residue formula comes from integrating $f^{(n)}$ repeatedly; it was skipped in the lecture

2.4 High Dimensional Taylor Expansion

- For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ we approximate by a multivariate polynomial and fix the coefficients by matching derivatives at $\mathbf{x}^*$ in the same way. To second order,

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*)$$

- $\mathbf{H}$ is the *Hessian*, the matrix of second derivatives

$$H_{ij} = \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j}$$

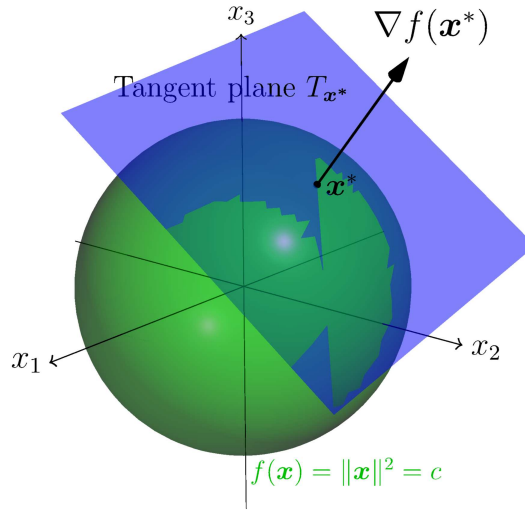
- For functions with continuous second derivatives the order of differentiation does not matter, so the Hessian is symmetric, $\mathbf{H} = \mathbf{H}^\top$
- Do not confuse the Hessian (second derivatives of a scalar function) with the Jacobian (first derivatives of a vector-valued function)

2.5 The Tangent Plane and Steepest Ascent

- For a smooth function the set $f(\mathbf{x}) = c$ is an $(n-1)$ -dimensional surface, the high dimensional analogue of a contour line
- The *tangent plane* at $\mathbf{x}^*$ is the plane that touches this surface at $\mathbf{x}^*$. Moving within it changes f only at second order, so it is the set of points with

$$(\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) = 0$$

that is, the tangent plane is orthogonal to the gradient



- Moving a small distance ϵ in a unit direction $\mathbf{u}$,

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}^*) + O(\epsilon^2)$$

- By the Cauchy-Schwarz inequality $\mathbf{u}^\top \nabla f \leq \|\nabla f\|$, with equality when $\mathbf{u}$ points along ∇f . So
 - $\nabla f(\mathbf{x}^*)$ is the direction of *steepest ascent*
 - $-\nabla f(\mathbf{x}^*)$ is the direction of *steepest descent*
- This is only a local statement: it holds while the $O(\epsilon^2)$ term is negligible

2.6 The Hessian at Stationary Points

- At a *stationary point*, $\nabla f(\mathbf{x}^*) = \mathbf{0}$, the first order term vanishes and

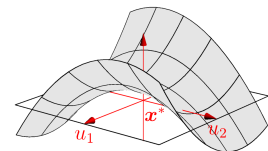
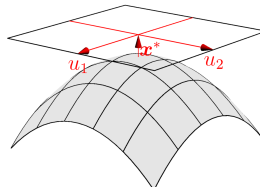
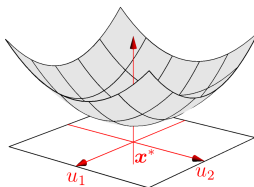
$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \frac{1}{2} \epsilon^2 \mathbf{u}^\top \mathbf{H} \mathbf{u} + O(\epsilon^3)$$

so the sign of $\mathbf{u}^\top \mathbf{H} \mathbf{u}$ decides whether f goes up or down

- $\mathbf{H}$ is real and symmetric, so it has real eigenvalues λ_i and orthonormal eigenvectors $\mathbf{v}_i$ ($\mathbf{H} \mathbf{v}_i = \lambda_i \mathbf{v}_i$, $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$)
- Writing any direction as $\mathbf{u} = \sum_i c_i \mathbf{v}_i$,

$$\mathbf{u}^\top \mathbf{H} \mathbf{u} = \left(\sum_i c_i \mathbf{v}_i^\top \right) \left(\sum_j c_j \lambda_j \mathbf{v}_j \right) = \sum_{i,j} \lambda_j c_i c_j \mathbf{v}_i^\top \mathbf{v}_j = \sum_i \lambda_i c_i^2$$

- Hence
 - all $\lambda_i > 0$: $\mathbf{H}$ is *positive definite*, $\mathbf{H} \succ 0$, and $\mathbf{x}^*$ is a local minimum
 - all $\lambda_i < 0$: $\mathbf{H}$ is *negative definite*, $\mathbf{H} \prec 0$, and $\mathbf{x}^*$ is a local maximum
 - eigenvalues of both signs: $\mathbf{x}^*$ is a saddle point
 - if some $\lambda_i = 0$ the second order test is inconclusive and higher order terms decide



2.7 Finding Optima

- To minimise a loss $L(\mathbf{w})$ we look for $\nabla L(\mathbf{w}^*) = \mathbf{0}$ with a positive definite Hessian
- Except for very simple losses there are many local minima, and $\nabla L = \mathbf{0}$ cannot be solved analytically
- Instead we iterate, using gradient descent

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - r \nabla L(\mathbf{w}^{(t)})$$

which moves in the locally steepest downhill direction. Steps shrink as the gradient shrinks near a minimum. Choosing r well is subtle; we return to this in the optimisation lecture

3 Exercises

3.1 Radius of convergence

- Write down the Taylor expansion of $\log(1+x)$ about $x=0$ up to the x^4 term. Without doing any further calculation, for what values of x must the series converge?

3.2 A bound from the residue

- Use the residue form of Taylor's expansion to show that $\cos(x) \geq 1 - x^2/2$ for all real x

3.3 Classifying a stationary point

- For $f(x, y) = x^2 + 3xy + y^2$ find the gradient and the Hessian. Show that the origin is the only stationary point and classify it

3.4 Quadratic functions

- For $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}$ with $\mathbf{A}$ symmetric, expand $f(\mathbf{x} + \epsilon \mathbf{u})$ exactly. Hence find the gradient and Hessian, and explain why the second order Taylor expansion of f is exact

4 Answers

4.1 Radius of convergence

- The derivatives are $f^{(k)}(x) = (-1)^{k-1}(k-1)!/(1+x)^k$, so

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

- $\log(1+z)$ is singular at $z = -1$, a distance 1 from the expansion point, so the series converges for $|x| < 1$

4.2 A bound from the residue

- With $f(x) = \cos(x)$ we have $f(0) = 1$, $f'(0) = 0$ and $f''(\theta) = -\cos(\theta)$. Stopping after two terms,

$$\cos(x) = 1 + 0 \cdot x - \frac{x^2}{2} \cos(\theta_x)$$

- Since $\cos(\theta_x) \leq 1$ and $x^2 \geq 0$, it follows that $\cos(x) \geq 1 - x^2/2$

4.3 Classifying a stationary point

- $\nabla f = (2x + 3y, 3x + 2y)^\top$. Setting this to zero gives a linear system with determinant $2 \times 2 - 3 \times 3 = -5 \neq 0$, so the only solution is $x = y = 0$
- The Hessian is constant,

$$\mathbf{H} = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$$

with eigenvalues 2 ± 3 , i.e. 5 and -1 , and eigenvectors $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$

- The eigenvalues have opposite signs, so the origin is a saddle point: f increases along $(1, 1)$ and decreases along $(1, -1)$

4.4 Quadratic functions

- Expanding,

$$\begin{aligned} f(\mathbf{x} + \epsilon \mathbf{u}) &= \frac{1}{2}(\mathbf{x} + \epsilon \mathbf{u})^\top \mathbf{A} (\mathbf{x} + \epsilon \mathbf{u}) - \mathbf{b}^\top (\mathbf{x} + \epsilon \mathbf{u}) \\ &= f(\mathbf{x}) + \epsilon \mathbf{u}^\top (\mathbf{A} \mathbf{x} - \mathbf{b}) + \frac{1}{2} \epsilon^2 \mathbf{u}^\top \mathbf{A} \mathbf{u} \end{aligned}$$

using $\mathbf{x}^\top \mathbf{A} \mathbf{u} = \mathbf{u}^\top \mathbf{A}^\top \mathbf{x} = \mathbf{u}^\top \mathbf{A} \mathbf{x}$ for symmetric $\mathbf{A}$

- Comparing with the Taylor expansion, $\nabla f = \mathbf{A} \mathbf{x} - \mathbf{b}$ and $\mathbf{H} = \mathbf{A}$
- There are no terms beyond ϵ^2 , so the second order expansion is exact. This is why quadratics are such a useful model of a loss function near its minimum